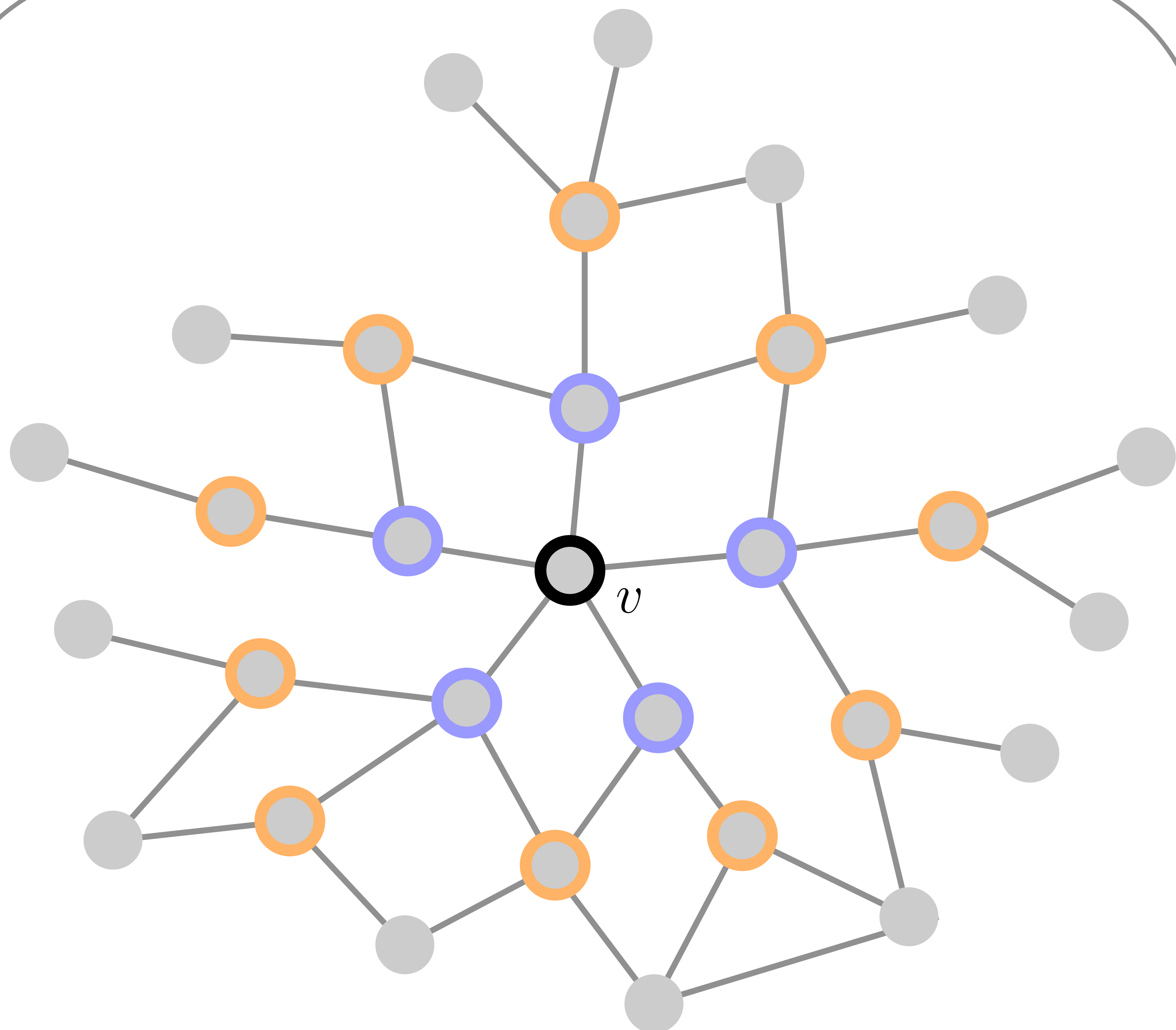
Embedding



$$g(v) = [\varphi(\mathcal{X}_v^1),\ \varphi(\mathcal{X}_v^2)] \in \mathbb{R}^8$$

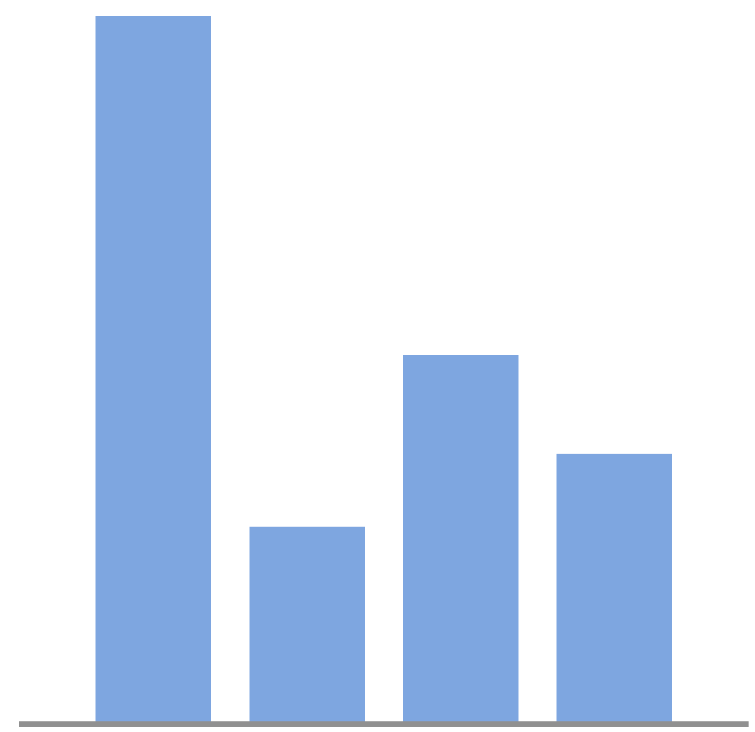
$$|\vec{x}_v\rangle \approx \begin{bmatrix} g^T(v) \\ \dots \end{bmatrix}$$

$$\frac{1}{N}\sum_{v=0}^{N-1} |v\rangle|0\rangle \rightarrow \frac{1}{N}\sum_{v=0}^{N-1} |v\rangle|\vec{x}_v\rangle$$

Classification

$$\frac{1}{N}\sum_{v=0}^{N-1} |v\rangle|\vec{x}_v\rangle|y_v\rangle|\vec{x}_{\text{test}}\rangle$$

After the swap test:



$\rightarrow \langle\sigma_z\sigma_z\rangle$